

MACHINE LEARNING FUNDAMENTALS SERIES

# Entropy for Data Science

## Quantifying Uncertainty, Surprise, and Information

**Source:** StatQuest with Josh Starmer — *Entropy (for data science) Clearly Explained!!!*

### 1 Visual Intuition: The Chicken Experiment

To understand **Entropy** intuitively, we first need to understand **Surprise** through a simple experiment. Imagine two types of chickens: **Orange** and **Blue**, organized into three distinct areas (A, B, and C).

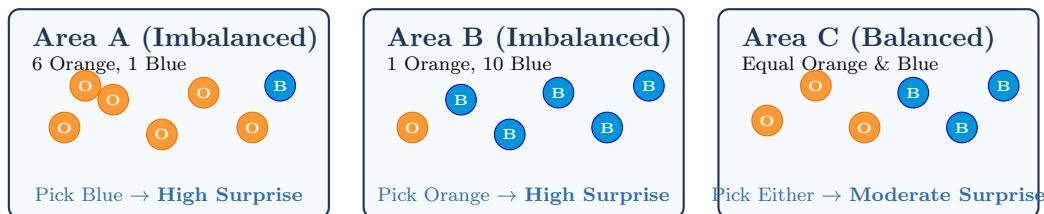


Figure 1: Visualizing Surprise across different class distributions.

- **Area A:** Orange is highly probable ( $P = 6/7$ ). Picking the rare Blue chicken ( $P = 1/7$ ) is **highly surprising**.
- **Area B:** Blue is highly probable ( $P = 10/11$ ). Picking an Orange chicken ( $P = 1/11$ ) is **highly surprising**.
- **Area C:** Both colors are equally likely ( $P = 0.5$ ). Picking either color is **equally, moderately surprising**.

**Core Intuition:** Surprise is inversely related to probability:

$$\text{High Probability } P(x) \implies \text{Low Surprise, Low Probability } P(x) \implies \text{High Surprise} \quad (1)$$

### 2 Quantifying Surprise

To model surprise mathematically, we need a function  $S(x)$  satisfying three properties:

1. If an event is certain ( $P(x) = 1$ ), surprise must be zero ( $S = 0$ ).
2. As  $P(x) \rightarrow 0$ , surprise grows ( $S(x) \rightarrow \infty$ ).
3. For independent events, total surprise is additive:  $S(x_1 \text{ and } x_2) = S(x_1) + S(x_2)$ .

Simple inversion  $1/P(x)$  fails property (1) because  $1/1 = 1 \neq 0$ . Taking the base-2 logarithm of the inverse probability satisfies all three properties cleanly:

#### Mathematical Definition of Surprise

For an event  $x$  with probability  $P(x)$ , **Surprise**  $S(x)$  (measured in *bits*) is defined as:

$$\text{Surprise}(x) = \log_2 \left( \frac{1}{P(x)} \right) \quad (2)$$

### 3 Definition and Meaning of Entropy

While **Surprise** measures the unexpectedness of a *single outcome*, **Entropy** quantifies the **expected (average) surprise per observation** across an entire probability distribution.

#### Definition of Entropy

**Entropy** ( $H$ ) is the **Expected Value** ( $\mathbb{E}$ ) of surprise over all possible outcomes of a random variable  $X$ :

$$H(X) = \mathbb{E}[\text{Surprise}] = \sum_x P(x) \times \log_2 \left( \frac{1}{P(x)} \right) \quad (3)$$

- **Low Entropy** ( $H \rightarrow 0$ ): High purity / predictability (one outcome dominates the population).
- **High Entropy** ( $H \rightarrow \text{Max}$ ): Maximum disorder / uncertainty (all outcomes are equally likely).

#### 4 Step-by-Step Derivation to Claude Shannon's Formula

Although the intuitive formula  $H(X) = \sum P(x) \log_2 \left( \frac{1}{P(x)} \right)$  clearly shows *Probability*  $\times$  *Surprise*, Claude Shannon originally published the formula in 1948 using a negative sign. Below is the step-by-step transformation:

##### Step 1: Start with the Intuitive Formula

$$H(X) = \sum_{i=1}^n P(x_i) \cdot \log_2 \left( \frac{1}{P(x_i)} \right) \quad (4)$$

##### Step 2: Apply Logarithmic Division Rule

Using the quotient rule for logarithms ( $\log_b \left( \frac{a}{c} \right) = \log_b(a) - \log_b(c)$ ) with  $a = 1$  and  $c = P(x_i)$ :

$$\log_2 \left( \frac{1}{P(x_i)} \right) = \log_2(1) - \log_2(P(x_i)) \quad (5)$$

##### Step 3: Simplify $\log_2(1) = 0$

Since  $\log_b(1) = 0$  for any log base  $b$ :

$$\log_2 \left( \frac{1}{P(x_i)} \right) = 0 - \log_2(P(x_i)) = -\log_2(P(x_i)) \quad (6)$$

##### Step 4: Substitute Back into the Expectation Sum

Replace  $\log_2 \left( \frac{1}{P(x_i)} \right)$  with  $-\log_2(P(x_i))$  inside the summation:

$$H(X) = \sum_{i=1}^n P(x_i) \cdot [-\log_2(P(x_i))] = \sum_{i=1}^n -P(x_i) \log_2(P(x_i)) \quad (7)$$

##### Step 5: Factor Out the Negative Sign (Classic Shannon Formula)

Factoring the constant negative sign outside the summation yields the classic formula published by Claude Shannon (1948):

##### Classic Claude Shannon Entropy Formula (1948)

$$H(X) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i) \quad (8)$$

#### 5 Numerical Example: Calculating Entropy for Areas A, B, and C

Using the chicken experiment from the video, let us compute the exact entropy values:

Table 1: Step-by-step Entropy Computation for Areas A, B, and C

| Area   | $P(\text{Orange})$   | $P(\text{Blue})$      | Surprise(O) | Surprise(B) | Calculation                                  | Entropy $H(X)$   |
|--------|----------------------|-----------------------|-------------|-------------|----------------------------------------------|------------------|
| Area A | $6/7 \approx 0.857$  | $1/7 \approx 0.143$   | 0.222 bits  | 2.807 bits  | $\frac{6}{7}(0.222) + \frac{1}{7}(2.807)$    | <b>0.59 bits</b> |
| Area B | $1/11 \approx 0.091$ | $10/11 \approx 0.909$ | 3.459 bits  | 0.138 bits  | $\frac{1}{11}(3.459) + \frac{10}{11}(0.138)$ | <b>0.44 bits</b> |
| Area C | $1/2 = 0.500$        | $1/2 = 0.500$         | 1.000 bit   | 1.000 bit   | $0.5(1.00) + 0.5(1.00)$                      | <b>1.00 bit</b>  |

#### 6 Key Applications in Machine Learning

- **Decision Trees (Information Gain)**: Used to select optimal feature splits by maximizing the drop in entropy ( $\text{Information Gain} = H_{\text{parent}} - H_{\text{children}}$ ).
- **Cross-Entropy Loss**: Measures the divergence between true label distribution  $P$  and predicted probabilities  $Q$  in classification neural networks.
- **Kullback-Leibler (KL) Divergence / Relative Entropy**: Quantifies information loss when approximating one distribution with another (found in t-SNE, UMAP, VAEs).